

Collections /Data Structures & Algorithms Course Content

❖ Pre requisite: Core Java knowledge

This Course is divided into 2 parts

DSA =

Complete Collections workshop + Time /Space complexity/
Algorithms/Sorting/Searching/Dynamic Programming/Pattern
Matching/Matrix/Graph

❖ Complete Collections workshop - Course Content

- Need for Collection Framework.
- Collection hierarchy
- Collection vs Collections
- Internal Implementation of List ✓
- Internal Implementation of Set
- Internal Implementation of Map
- How to create custom collection
- List
 - Array List ✓
 - Vector ✓
 - Linked List ✓
- Copy on Write Array List ✓
- Set
 - Hash Set ✓
 - Linked Hash Set ✓
 - Copy on write Array Set ✓
- Map
 - Hash Map ✓
 - Linked Hash Map ✓
 - Concurrent hash Map ✓
 - How to sort hashmap by values ? ✓
 - How to sort hashmap by keys ✓
 - Why to over ride equals and hash code method in hashmap ? ✓
- What is iterator and how to use the iterator ? ✓
- What is List Iterator and how to use it ? ✓
- Difference between iterator and List iterator ? ✓
- What is concurrent modification exception and how to resolve it? ✓
- Concurrency vs Synchronization ? ✓
- Copy on write Array list , Copy on write array Set , Concurrent hash map
- How to create Custom collection (Custom Array List) ? ✓
- Linked List
 - Introduction ✓
 - Insert (Head/Tail/Middle) ✓
 - <https://www.hackerrank.com/challenges/insert-a-node-at-the-head-of-a-linked-list/problem>
 - <https://www.hackerrank.com/challenges/insert-a-node-at-a-specific-position-in-a-linked-list/problem>
 - <https://www.hackerrank.com/challenges/insert-a-node-at-the-tail-of-a-linked-list/problem>
 - Delete (Head/Tail/Middle) ✓
 - <https://www.hackerrank.com/challenges/delete-a-node-from-a-linked-list/problem>
 - Print: ✓
 - <https://www.hackerrank.com/challenges/print-the-elements-of-a-linked-list-in-reverse/problem>
 - <https://www.hackerrank.com/challenges/print-the-elements-of-a-linked-list/problem>
 - <https://www.hackerrank.com/challenges/get-the-value-of-the-node-at-a-specific-position-from-the-tail/problem>
- Stack:
 - What is Stack ? ✓
 - Basic Stack properties (Push, Pop, Peek) ✓
 - <https://www.hackerrank.com/challenges/java-stack/problem>
 - Stack Implementation using Array ✓
 - Stack Implementation using Linked List ✓

- Basic Stack properties (Push, Pop, Peek) ✓
- <https://www.hackerrank.com/challenges/java-stack/problem>
- Stack Implementation using Array ✓
- Stack Implementation using Linked List ✓
- Stack implementation using Queue ✓
- **Queue:**
 - What is Queue ? ✓
 - Basic Properties of Queue (Peek, poll) ✓
 - What is D Queue, Blocking Queue, Priority Queue ? ✓
 - What are the difference between Linked Blocking Queue verses Array Blocking Queue ? ✓
 - Queue Implementation using Array ✓
 - Queue Implementation using Linked List ✓
 - Queue implementation using Stack
- **Trees**
 - Basic concepts
 - In order traversal: <https://www.hackerrank.com/challenges/tree-inorder-traversal/problem>
 - Post order traversal: <https://www.hackerrank.com/challenges/tree-postorder-traversal>
 - Preorder traversal: <https://www.hackerrank.com/challenges/tree-preorder-traversal/problem>
- **Tree Set**
- **Tree Map**

➤ Time/Space Complexity/Algorithms Topics/ Sorting/ Searching/ Dynamic Programming/Matrix/Graph- Course Content

- **What is Time complexity ?**
- **What is space complexity ?**
- **Sorting**
 - **Bubble sort**
<https://www.hackerrank.com/contests/ashok-it-dsa/challenges/bubble-sort-implementation>
 - **Insertion sort**
<https://www.hackerrank.com/contests/ashok-it-dsa/challenges/insertion-sort-implementation>
<https://www.hackerrank.com/challenges/insertionsort1/problem>
<https://www.hackerrank.com/challenges/insertionsort2/problem>
<https://www.hackerrank.com/challenges/insertion-sort/problem>
 - **Selection sort:**
<https://www.hackerrank.com/contests/ashok-it-dsa/challenges/selection-sort-implementation>
 - **Two way Merging/Three way merging**
 - **Merge sort & Merge sorted array:** <https://leetcode.com/problems/merge-sorted-array/>
 - **Quick Sort**
 - **What is Max Heap ?**
 - **What is Min Heap ?**
 - **What is Heap Sort?**
 - Convert the given array into Max Heap ?
 - Convert the given array into Min Heap ?
 - Adding an element to Max Heap or Min Heap ?
- **Graphs**
 - **Basic Concept and How to Construct the Graph in Java ?**
 - **BFS**
 - **DFS**
 - **Shortest Path from source to destination (Dijkstra's algorithm)**
- **Dynamic Programming**
 - **Fundamental OF DP**
 - **Fibonacci Series Problem**
 - **Recursion verses Dynamic Programming**
 - **Coin Change Problem**
- **Matrix:**
 - **Matrix Representation**
 - **Addition of 2 matrix**
 - **Multiplication of 2 matrix**
 - **Check whether the given matrix is symmetric or not ?**

- Printing the Matrix
- Transpose of a Matrix

➤ Pattern Printing

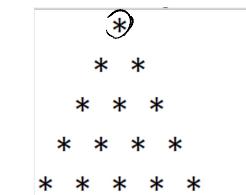
<https://www.hackerrank.com/challenges/staircase/problem?isFullScreen=true>

Triangle Pattern

- Pattern Programs for the below patterns



Pattern 2



Pattern 3



➤ Programmatic Questions

- Fundamentals of Array
- Min and Max element in the array
- Unique Elements of an array
- Use Recursion to reverse the array
- Duplicate elements of an array
- Search for the missing number
- Factorial of a number
- Nth Fibonacci number
- Prime or not:
- Basic Implementation
- <https://www.hackerrank.com/challenges/simple-array-sum/problem>
- Reverse of an Array: <https://www.hackerrank.com/challenges/arrays-ds/problem?isFullScreen=true>
- LCM of N numbers: <https://www.hackerrank.com/contests/crescent-contest/challenges/find-the-lcm>
- Simple Array Sum: <https://www.hackerrank.com/challenges/simple-array-sum/problem>
- Sum of N numbers: <https://www.hackerrank.com/contests/placement-prep-testing/challenges/sum-of-n-numbers>
- What is the simplest way to identify the array is in sorted manner with less complexity ?
- Write a program to reverse the array without using extra memory ?
- Swap two numbers without using the third variable?
- Pair difference: www.interviewbit.com/problems/diffk/
- Linear & Binary Search: <https://leetcode.com/problems/two-sum/>
- Square Root: <https://www.hackerrank.com/contests/ashok-it-dsa/challenges/square-root-10>
- Compare two Linked list :

- <https://www.hackerrank.com/challenges/compare-two-linked-lists/problem>
- Merge 2 sorted Linked List
<https://www.hackerrank.com/challenges/merge-two-sorted-linked-lists/problem>
- Cyclic detection
<https://www.hackerrank.com/challenges/detect-whether-a-linked-list-contains-a-cycle/problem>
- Open and close bracket problem using Stack:
<https://www.hackerrank.com/challenges/java-stack/problem>
- Min and Max Element Present in a Binary Search Tree ?
- Print all leaves of a binary search tree?
- Height of A Tree: <https://www.hackerrank.com/challenges/tree-height-of-a-binary-tree/problem>
- Fill Depth: <https://www.hackerrank.com/contests/ashok-it-dsa/challenges/find-depth>
Sum OF all the Elements Present in a Tree
- Level order traversal : <https://www.hackerrank.com/challenges/30-binary-trees/problem>
- Vertical order traversal : <https://leetcode.com/problems/vertical-order-traversal-of-a-binary-tree/>
- Distance Between 2 Nodes in BST : <https://leetcode.com/problems/minimum-distance-between-bst-nodes/>
- Left-View of A Tree : <https://www.geeksforgeeks.org/problems/left-view-of-binary-tree/1>
- Check Whether the given tree is BST or not ?
- Write a program to print the mirror image of a Tree ?
https://www.codingninjas.com/studio/problems/second-largest-element-in-the-array_873375
<https://practice.geeksforgeeks.org/problems/merge-two-sorted-arrays-1587115620/1>

➤ **Fee :**

Only complete **Collection workshop : 3000 + Notes and video access up to 6 months**
DSA:

Only collections (Recordings to 6 months access): 3000

Only DSA: 8000

Collections class back up videos live classes for AD5A: 10,000 /-for Live classes and complete course videos up to 1 Year access

Timing: 7 AM to 8 AM (Mon to Friday)

➤ **Duration:**

30 to 35 days for Complete Collection framework
DSA: 50 to 60 Days